

Capstone Project Submission Package

Project Name: Sentinel-FinAI

1. Working AI Agent

Source: backend/agents/adaptive_agent_v7.py

A fully integrated AI agent utilizing LangGraph for workflow management, ChatOpenAI (GPT-4o) for reasoning, and ChromaDB for RAG-based knowledge retrieval. Features include adaptive memory and custom financial tools.

2. Problem Framing Document

Source: Problem_Framing_Sentinel_FinAI.pdf

Comprehensive 1-page document outlining the business problem, target audience, technical solution, and success metrics for the banking advisory agent.

3. Demo Script (Forced Interactions)

Source: backend/phase-8.md

A set of 5-10 validated scenarios covering Identity (Greeting), Accuracy (RAG), Safety (Refusal), Tool Use (Calculations), and Resilience (Gibberish handling).

4. Evaluation Report

Source: backend/logs/engineering_review_final.json

Automated engineering report capturing average latency, response accuracy, safety adherence, and root cause analysis of system failures.

5. Engineering & Product Justification

Source: `backend/templates/prompt_registry.json`

Justification for using a versioned Prompt Registry (v7.1-v7.3) for stability, early-exit validation logic for resilience, and safety-first prompting to mitigate financial liability.